KEY PROJECT INFORMATION & PROJECT DESIGN DOCUMENT (PDD)

PUBLICATION DATE **14.10.2020**

VERSION **v. 1.2**

RELATED SUPPORT

- TEMPLATE GUIDE Key Project Information & Project Design Document v.1.2

This document contains the following Sections

KEY PROJECT INFORMATION

GS ID of Project	GS11423
Title of Project	160MW Balaoi & Caunayan Wind Energy Project
Time of First Submission Date	20/10/2021
Date of Design Certification	15/11/2023
Version number of the PDD	1.5
Completion date of version	19/02/2024
Project Developer	Bayog Wind Power Corp.
Project Representative	Mark Anthony Tagum
Project Participants and any communities involved	Bayog Wind Power Corp.
Host Country (ies)	Philippines
Activity Requirements applied	☐ Community Services Activities ☒ Renewable Energy Activities ☐ Land Use and Forestry Activities/Risks & Capacities ☐ N/A
Scale of the project activity	☐ Micro scale ☐ Small Scale ☒ Large Scale
Other Requirements applied	NA
Methodology (ies) applied and version number	ACM0002: Grid-connected electricity generation from renewable sources – Version 21.0
Product Requirements applied	☒ GHG Emissions Reduction & Sequestration ☐ Renewable Energy Label ☐ N/A
Project Cycle:	☐ Regular ☒ Retroactive

Table 1 – Estimated Sustainable Development Contributions

Sustainable Development Goals Targeted	SDG Impact (defined in B.6)	Estimated Annual Average	Units or Products
13 Climate Action (mandatory)	Emission Reductions	363,894	tCO2e/year VERs
7 Affordable and Clean Energy	Ensure access to affordable, reliable, sustainable and modern energy for all	532,300	MWh/year (of renewable energy generated)
8 Decent Work and Economic Growth:	Promote sustained, inclusive and sustainable economic growth, full and productive employment and decent work for all	10	Employees

SECTION A. DESCRIPTION OF PROJECT

A.1 Purpose and general description of project

The 160MW Balaoi & Caunayan Wind energy project is located in Barangays Balaoi and Caunayan, Municipality of Pagudpud, Province of Ilocos Norte, Philippines. It is being developed by Bayog Wind Power Corp. The total installed capacity of the wind power project will be 160MW. The project site is divided into 3 distinct areas/phases of Barangays Balaoi and Caunayan but project implementation will happen simultaneously and will have one substation. It has been estimated that the operating lifetime of the proposed project is expected to be 20 years.

The project will install 32 units of wind turbines each with an individual capacity of 5MW thus, resulting in total installed capacity of 160MW. A new 15km 115kV interconnection line will be constructed by Bayog Wind Power Corp. to connect the project to an existing double circuit transmission line of North Luzon Renewables in Bangui which is already connected to NGCP's substation in Laoag city thus, connecting the project to NGCP's transmission system. The Bangui to Laoag transmission line segment is approximately 42km, thus total interconnection line to connect the project to the grid is 57km. The project is expected to generate 532,300 MWh of electricity annually and will be sold to Wholesale Electricity Spot Market.

The project aims to provide clean, renewable, reliable, and sustainable energy to the grid of the Philippines. It will help in meeting the rising demand for energy to bolster nation's expanding economy. By utilizing wind energy, an abundant renewable energy resource in the country, the project will not just provide clean energy to the grid but will also reduce nation's high dependence on imported fossil fuels. Besides this, the project will also contribute towards bridging the gap between demand and supply of power and therefore, will strengthen national energy security. The project is expected to reduce 363,894 tCO2e of emissions annually.

The baseline scenario of the project involves grid connected electricity generation in the Luzon grid of Philippines. By using wind energy, the project displaces CO2 emissions generated by fossil-fuel fired power plants. In the absence of the proposed project, the electricity would have been generated by existing fossil-fuel fired power plants along with addition of new fossil-fuel based power plants.

A.1.1. Eligibility of the project under Gold Standard

The project activity meets the eligibility criteria as per section 3.1.1 of GS4GG Principles & Requirements document as described below:

S.NO.	Eligibility Conditions	Justification
1.	Types of Project: Eligible projects shall include physical action/implementation on the ground. Pre-identified eligible project types are identified in the Eligibility Principles and Requirements section.	The proposed project activity will use wind energy, a renewable source of energy to generate electricity which will be delivered to the grid through Wholesale Electricity Spot Market. The project employs CDM approved methodology ACM0002, Version 21.0, an eligible methodology under Gold Standard for Global Goals.
2.	Location of Project: Projects may be located in any part of the world.	The project activity is located in Barangays Balaoi and Caunayan, Pagudpud, Ilocos Norte, Philippines. It is further detailed in section A.2.
3.	Project Area, Project Boundary and Scale: The Project Area and Project	The project area and boundary are defined in line with the applicable methodology ACM0002, Version 21.0.

	Boundary shall be defined. Projects may be developed at any scale although certain rules, requirements and limitations may apply under specific Activity Requirements, Impact Quantification Methodologies and Products Requirements. In order to avoid double counting the Project shall not be included in any other voluntary or compliance standards programme unless approved by Gold Standard (for example through dual certification). Also, if the Project Area overlaps with that of another Gold Standard or other voluntary or compliance standard programme of a similar nature, the project shall demonstrate that there is no double counting of impacts at design and performance certification (for example use of similar technology or practices through which the potential arises for double counting or misestimation of impacts amongst projects).	According to the applied CDM methodology, the spatial extent of the project boundary incorporates 160MW Balaoi & Caunayan Wind Energy Project, substation, all the power plants connected physically to the national grid to which the proposed project will also be connected. The project boundary diagram is given in section B.3 of this PDD. The project activity is an installation of a Greenfield power project with total installed capacity of 160MW which is more than 15MW and hence, is categorized as a large-scale power plant. The project is not included in any other voluntary or compliance standards programme. Furthermore, the project's area does not overlap with any other projects, belonging to Gold Standard or any other Voluntary or compliance standard programme. Hence there is no question of double counting. The Project Developer also confirms that the host country, region, locality or state does not have an emission reduction cap enforced OR have the possibility to trade emissions that include the scope of the proposed project. International Carbon Action Partnership (ICAP) is an international forum for governments and public authorities that have implemented or are planning to implement emissions trading systems (ETS). ICAP facilitates cooperation between countries, sub-national jurisdictions, and supranational institutions that have established or are actively pursuing carbon markets through mandatory cap and trade systems. According to ICAP, the status of implementing a national Cap-and-Trade System in the host country (Philippines) is under Consideration. "Low Carbon Economy Act" House Bill (HB) No. 2184 which includes provisions for a domestic cap-and-trade system was conditionally approved by the Committee on Climate Change of the Philippine House of Representatives in early 2020. The Bill shall impose a cap on Industrial and Commercial sectors. But there is no specification on the implementation status of the Act in the Bill. However, it has been pending with the Technical Working Committee since then. As per ICAP, an updated version of this bill is expected in this year but none of the three institutions involved such as: Department of Environment and Natural Resources, Department of Trade and Industry and House of Representatives Committee on Climate Change have neither come up with any further update on this Bill nor have given any clear intimation regarding any update so far.
4.	Host Country Requirements: Projects shall be in compliance with applicable Host Country's legal, environmental, ecological and social regulations.	The project is in compliance with applicable host country's legal, environmental, ecological and social regulations as it has obtained necessary certificates/agreements such as Environmental Compliance Certificate and Forest Land Use Agreement from the Department of Environment and Natural Resources, Philippines, Certification of Registration from the Board of

		Investment (Department of Trade and Industry), Philippines and Wind Energy Certificate of Registration from the Department of Energy, Philippines.
5.	Contact Details: As part of the Project Documentation the Project Developer shall provide (i) name and (ii) contact details of all Project Participants; AND in case of an organisation (iii) the legal registration details and (iv) documentation by the governing jurisdiction that proves that the entity is in good standing (defined as being a legal or other appropriate entity registered in or allowed to operate within the required jurisdiction and with no evidence of insolvency or legal/criminal notices placed against it or any of its Directors). Gold0020Standard retains the right (at its own discretion) to refuse use of the Standard where reputational concerns are highlighted.	All such details are mentioned in Appendix 1 of the PDD. The project had obtained Certificate of Good Standing from the Board of Investments, Philippines which clearly denotes that BWPC exists, lawful business owner as it complies to all the laws and is allowed to conduct business, thus proving that entity is of good standing.
6.	Legal Ownership: Full and uncontested legal ownership of any Products that are generated under Gold Standard Certification, (for example carbon credits) shall be demonstrated. Where such ownership is transferred from project beneficiaries this must be demonstrated transparently and with full, prior and informed consent (FPIC). Note that for certain Project types there is a requirement for full and uncontested legal land title/tenure to be demonstrated. These are contained within specific Activity or Product Requirements. All projects shall immediately report to Gold Standard any land title/tenure disputes arising.	The project activity is being undertaken by Bayog Wind Power Corp. which is the sole owner of the project activity and any Products generated under Gold Standard Certification. Supporting documents such as Certification of Registration from the Board of Investment (Department of Trade and Industry), Philippines and Wind Energy Certificate of Registration from the Department of Energy, Philippines demonstrating the same are available for verification.
7.	Other Rights: As well as legal title and ownership, the Project Developer shall also demonstrate where required uncontested legal rights and/or permissions concerning changes in use of other resources required to service the Project (for example, access rights, water rights etc.). Any known disputes or contested rights must be declared immediately to Gold Standard by the	Bayog Wind Power Corp. owns all the legal rights and is the rightful owner of the project. It has already obtained all the necessary permits/consents such as Wind Energy Certificate of Registration, Environmental Compliance Certificate and Forest Land Use Agreement from the various concerned government departments that are required to develop

	Project Developer and resolved prior to further project implementation in affected areas.	and operate a wind energy project in the Philippines.
8.	Official Development Assistance (ODA) Declaration: All Project Developers applying for project activities located in a country named by the OECD Development Assistance Committee's ODA recipient list and seeking Gold Standard Certification for carbon credits shall declare the Official Development Assistance (ODA) support. The Project Developer shall follow the GHG Emissions Reduction & Sequestration Product Requirements and submit the declaration at the time of Design Certification.	Philippines is in the DAC list of ODA recipients, 2020 under the category: Low Middle-Income Countries and Territories. The project does not receive any public funding or ODA support and an ODA Declaration signed by the project developer confirming the same is available.

The project meets the specific eligibility criteria as per sections 2 and 3 of "GS4GG Renewable Energy Activity Requirements", version 1.4, as described below:

S.No.	Eligibility Conditions	Justification
1.	Projects shall generate and deliver energy services (e.g. mechanical work/electricity/heat) from non-fossil and renewable energy sources.	The project will generate electricity from wind, a non-fossil, renewable source of energy.
2.	Projects shall comprise of renewable energy generation units, such as photovoltaic, tidal/wave, wind, hydro, geothermal, waste to energy and renewable biomass, that are • Supplying energy to a national or a regional grid; OR • Supplying energy to an identified consumer facility via national/regional grid through a contractual agreement such as wheeling	The project will generate electricity by using wind energy and electricity will be supplied to the grid through Wholesale Electricity Spot Market. The Wholesale Electricity Spot Market (WESM) is a platform for trading electricity as a commodity. It is governed by the Philippine Electricity Market Corporation (PEM), a non-stock, non-profit corporation. It sets the WESM rules for the Market Operator, System Operator, WESM Participants and PEM board. The WESM rules act complimentary to the Grid and Distribution Code of the Philippines and its aim is to establish a competitive, efficient, transparent and reliable market for electricity trading.
3.	Any Project supplying electricity to a mini-grid shall refer to Community Services Activity Requirements.	The condition is not applicable as the project activity is not supplying to a mini-grid.

4.	Projects generating on-site energy for captive consumption at an industrial facility shall refer to the requirements in this document.	The condition is not applicable as the project does not generate on-site energy for captive consumption.
5.	New Gold Standard Verified Emission Reductions (GS VER) or Gold Standard labels for Certified Emission Reductions (GS CER), Renewable Energy projects connected to national or a regional electricity grid must be located in a; a. Least Developed Country (LDC), Small Island Developing State (SIDS) or a Land Locked Developing Country (LLDC) or b. Low Income and Low Middle-income country where the penetration level of the proposed Renewable Energy Technology type is less than 5% of the total grid installed capacity, at the time of the first submission to preliminary review. Renewable Energy projects connected to national or a regional electricity grid are ineligible for GS VERs/CERs, if located in: • An upper Middle-Income Country or High-Income Country or • SIDS and LLDC, defined as a High-Income Country	The project activity is located in the Philippines which is a Low-Middle income country. As per the latest government data available, the ratio of total installed capacity of Wind technology (443MW) to the total installed grid capacity (26,286MW) is 1.6% and hence, is less than 5%[1]. Therefore, the project activity is eligible.
6.	Grid Connected off-shore wind projects and waste to energy projects that involve utilization of landfill gas/biogas to electricity generation with or without thermal energy production are exempted from eligibility requirement outlined above.	The condition is not applicable as the project is an on-shore wind energy project.

Section 3: General Eligibility Criteria

- **Project Type:** As mentioned above, the project belongs to a pre-identified eligible project type. Moreover, the project is located in the Philippines, classified as a Low-Middle Income Country by the World Bank according to their latest classification of countries as per income level. The ratio of total installed capacity of Wind technology (443MW) to the total installed grid capacity (26,286MW) is 1.6%, which is less than

[1] https://agep.aseanenergy.org/wp-content/uploads/2022/03/The-Philippines-CP2022.pdf

5% as required by Paragraph 2.1.3 of section 2 of Renewable Energy Activity Requirements, version 1.4.Thus, the project is an eligible project type.

- **Location of project:** The project activity is located in the Barangays Balaoi and Caunayan, Municipality of Pagudpud, Province of Ilocos Norte, Philippines.
- **Project area, boundary and scale:** The project area and boundary are defined in line with the applicable methodology ACM0002, Version:21.0. According to the applied CDM methodology, the spatial extent of the project boundary includes the project boundary incorporates 160MW Balaoi & Caunayan Wind Energy Project, substation, all power plants connected physically to the national grid to which the proposed project will also be connected. The project boundary diagram is given in Section B.3 of this PDD.

  The project activity is an installation of a Greenfield power project with total installed capacity of 160MW which is more than 15MW and hence, is categorized as a large-scale power plant.
- **Suppressed Demand:** The project activity does not account for suppressed demand scenario while establishing baseline.
- **Stacking:** The project will only claim for GSVERs and therefore, will not be stacking.

A.1.2. Legal ownership of products generated by the project and legal rights to alter use of resources required to service the project

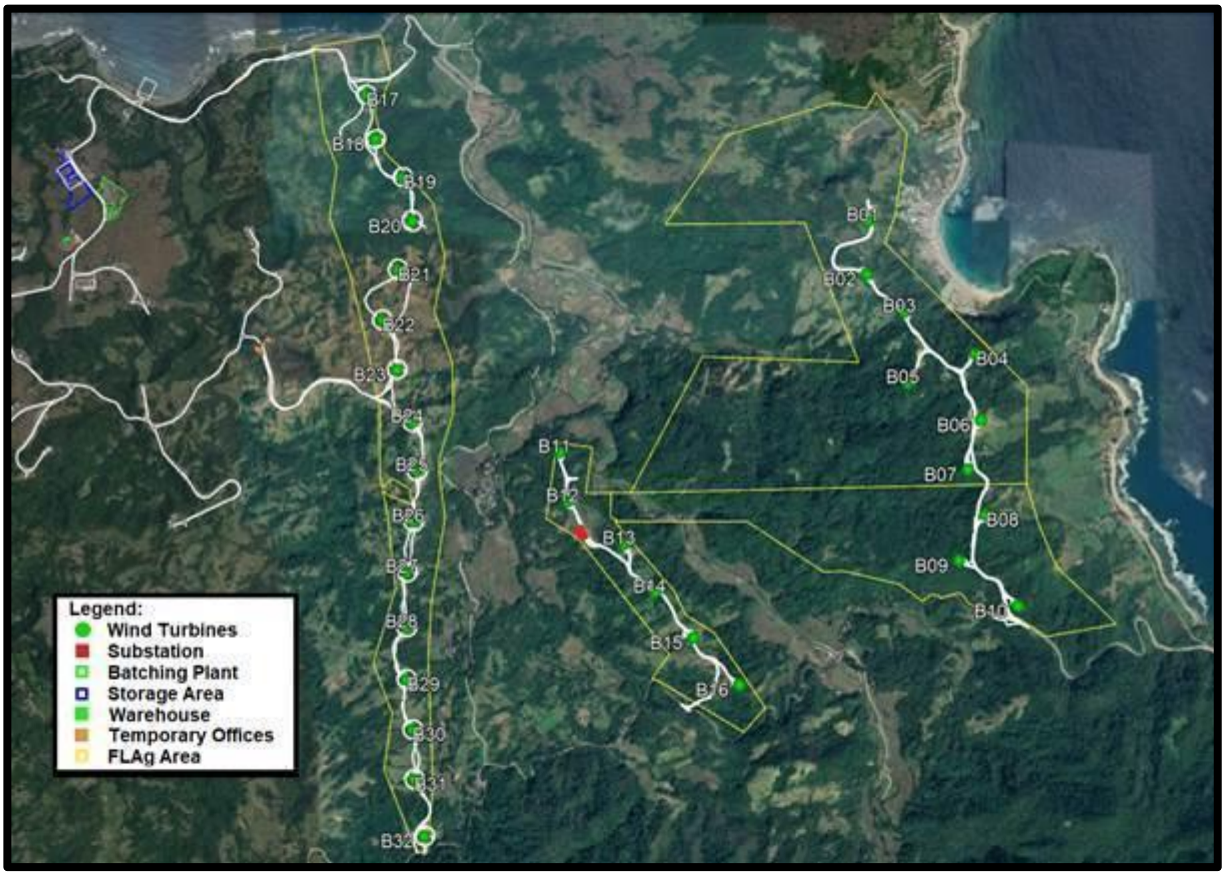
Bayog Wind Power Corp. is the legal owner of the project and will be responsible for handling each and every step of the process related to acquiring Gold Standard Certification and managing the same throughout the whole crediting period of the project.

Bayog Wind Power Corp. has the legal ownership of the carbon credits that will be generated by the project activity throughout the crediting period of the project and the company is entitled to make any change in the use of resources required to service the project at any stage of the project cycle.

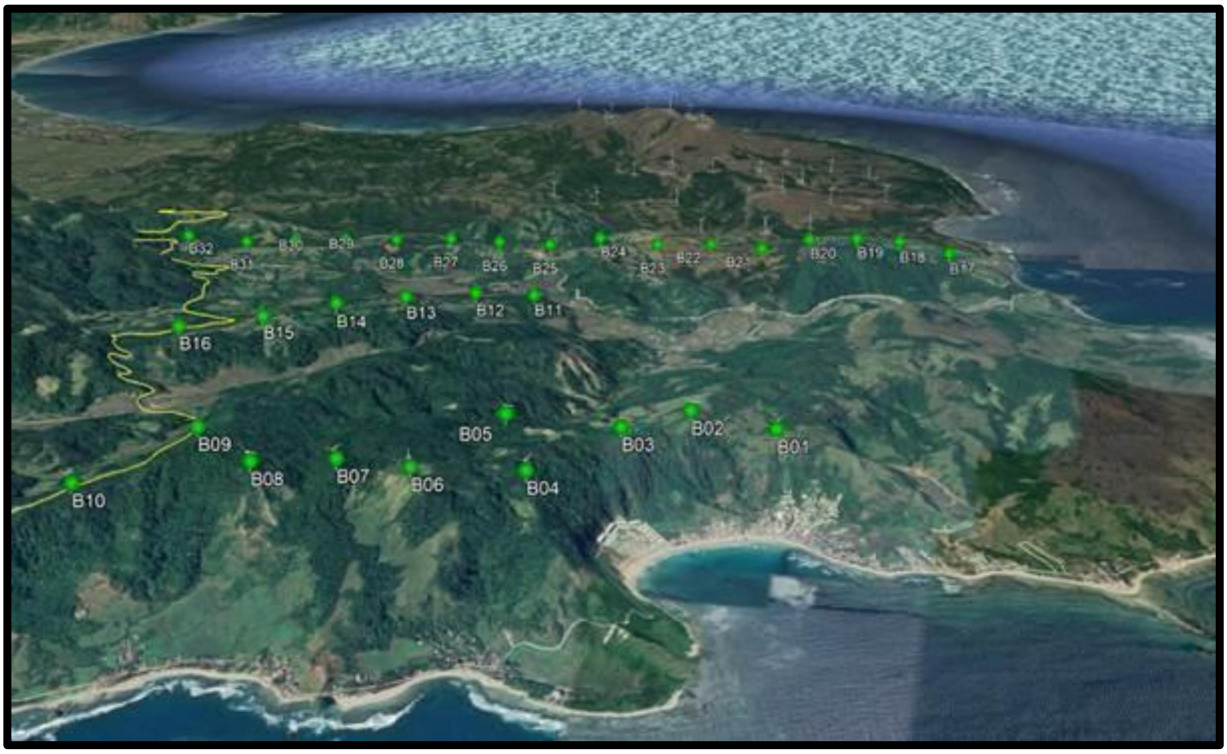
A.2 Location of project

Barangays Balaoi and Caunayan, Municipality of Pagudpud, Province of Ilocos Norte, Philippines.

The project is located in the northern part of Luzon Island. After conducting various studies, the project site was confirmed to be amongst one of the areas with best wind resources in the country. The project site layout and WTG location coordinates table have been attached below.



Project Site Layout Map



Project Site Layout Map

WTG	Turbine Type	Hub Height (m)	Coordinates (WGS84 UTM51)	
			X [m]	Y [m]
B01	SG5.0-132	84.0	273634	2060648
B02	SG5.0-132	84.0	273614	2060336
B03	SG5.0-132	84.0	273824	2060117
B04	SG5.0-132	84.0	274227	2059874
B05	SG5.0-132	84.0	273833	2059710
B06	SG5.0-132	84.0	274255	2059498
B07	SG5.0-132	84.0	274181	2059217
B08	SG5.0-132	84.0	274274	2058959
B09	SG5.0-132	84.0	274109	2058703
B10	SG5.0-132	84.0	274462	2058435
B11	SG5.0-145	107.5	271833	2059346
B12	SG5.0-145	107.5	271877	2059061
B13	SG5.0-145	107.5	272203	2058814
B14	SG5.0-145	107.5	272377	2058536

B15	SG5.0-145	107.5	272597	2058273
B16	SG5.0-145	107.5	272857	2058003
B17	SG5.0-145	107.5	270729	2061430
B18	SG5.0-132	106.5	270795	2061153
B19	SG5.0-132	106.5	270957	2060922
B20	SG5.0-132	106.5	271008	2060671
B21	SG5.0-145	107.5	270909	2060409
B22	SG5.0-145	107.5	270813	2060121
B23	SG5.0-145	107.5	270898	2059832
B24	SG5.0-132	106.5	270991	2059527
B25	SG5.0-145	107.5	271007	2059248
B26	SG5.0-145	107.5	270982	2058957
B27	SG5.0-145	107.5	270943	2058666
B28	SG5.0-145	107.5	270933	2058346
B29	SG5.0-145	107.5	270929	2058045
B30	SG5.0-145	107.5	270958	2057753
B31	SG5.0-145	107.5	270963	2057462
B32	SG5.0-145	107.5	271024	2057143

WTG Location Coordinates

A.3 Technologies and/or measures

A 160MW grid-connected wind power plant will be constructed by Bayog Wind Power Corp. The project site is considered a suitable location for constructing a wind farm as per pre-engineering studies. The wind farm will consist of 32 turbines of 5MW capacity each, a 200MVA substation and a 15km 115kV interconnection line. Two types of wind turbines called SG5.0-132 and SG5.0-145 with 14 and 18 units respectively will be installed at the project site. The kinetic energy of the wind will turn the propeller-like blades around a rotor, connected to a generator which will produce electricity. A new 15km 115kV interconnection line will be constructed by Bayog Wind Power Corp. to connect the project to an existing double circuit transmission line of North Luzon Renewables in Bangui which is already connected to NGCP's substation in Laoag city thus, connecting the project to NGCP's transmission system. Thus, this total 57km direct interconnection line will supply electricity from the project to the grid. In the baseline scenario, equal amount of electricity would be supplied by the fossil-fuel based power plants connected to the grid. The average lifespan of the Turbine is 20 years which will substantially decrease the emissions.

Technological Description:

- **Wind Turbines: As already mentioned above, the project site will have two types of wind turbines installed as given below:**
  1. **SG5.0-132 WTG:** The SG 5.0-132 IA wind turbines correspond to a three-bladed wind-facing rotor type with a baseline rated power of 5.0 MW. These wind turbines have a rotor diameter of 132 m and baseline tower heights of 10 units is 84m and 4 units is 106.5m depending on the height of the tower. The SG 5.0-132 IA wind turbines are regulated by an independent pitch control system in each blade and have an active yaw system. The control system allows the wind turbine to be operated at variable speed, maximizing the power production at all operation regimes and minimizing the loads and noise.
  2. The SG 5.0-145 IIB wind turbines correspond to a three-bladed wind-facing rotor type with a baseline rated power of 5.0 MW. These wind turbines of 18 units have a rotor diameter of 145 m and baseline hub heights of 79.5m, 90m, 107.5m and 127.5 m depending on the height of the tower. The SG 5.0-145 IIB wind turbines are regulated by an independent pitch control system in each blade and have an active yaw system. The control system allows the wind turbine to be operated at variable speed, maximizing the power production at all operation regimes minimizing the loads and noise.

The following tables include key technical description of the respective wind turbine models:

Model Type: **SG5.0-132**	
Key Technical Parameter	**Value**
Wind Turbine	
Capacity	5MW
Rotor Diameter	132m
Tower Height	10 units with tower height of 84m and 4 units is 106.5m
Technical lifespan of the Wind	20

Turbine	
Manufacturer	Siemens Gamesa
Generator	
Type	Doubly-fed with coil rotor and slip rings
Maximum Power (kW)	5,150 (stator + rotor)
Voltage (Vac)	690
Frequency (Hz)	60
Transformer	
Type	Three-phase, dry-type encapsulated
Frequency (Hz)	60
Insulation Class	F or H
Model Type: **SG5.0-145**	
Key Technical Parameter	**Value**
Capacity	5MW
Rotor Diameter	145m
Tower Height	18 units with tower height of 107.5m
Technical Lifespan of the Wind Turbine	20
Manufacturer	Siemens Gamesa
Generator	
Type	Doubly-fed with coil rotor and slip rings
Maximum Power (kW)	5.150 (stator + rotor)
Voltage (Vac)	690
Frequency (Hz)	60
Transformer	
Type	Three-phase, dry-type encapsulated
Frequency (Hz)	60
Insulation Class	F or H

A.4 Scale of the project

A renewable energy project with a maximum output of 15MW (or an appropriate equivalent) is categorized as a small-scale project. (As per Renewable Energy Activity Requirements, version:1.4) A project with maximum output above 15MW is considered as a large-scale project.

The total installed capacity of project activity will be 160MW and hence, it will be a large-scale project.

A.5 Funding sources of project

The project will not receive any public funding or ODA support as declared in the ODA Declaration.

SECTION B. APPLICATION OF APPROVED GOLD STANDARD METHODOLOGY (IES) AND/OR DEMONSTRATION OF SDG CONTRIBUTIONS

B.1. Reference of approved methodology (ies)

Applied Methodology

- Large-scale Consolidated Methodology ACM0002, Version:21.0 – "Grid-connected electricity generation from renewable sources"

Related Tools

- Tool to calculate the emission factor for an electricity system, Version:07.0
- Tool for the demonstration and assessment of additionality, Version:07.0.0
- Tool for Common Practice, Version: 03.1
- Tool for Investment analysis, Version: 10.0
- Tool for Baseline, project and/or leakage emissions from electricity consumption and monitoring of electricity generation, Version: 03.0

B.2. Applicability of methodology (ies)

The proposed project is a large-scale grid-connected Greenfield renewable energy power plant.

Large-scale Consolidated Methodology ACM0002, Version:21.0 – "Grid-connected electricity generation from renewable sources" is applicable for the proposed project as per the scope defined in the methodology. A detailed discussion regarding applicability of ACM0002 methodology, Version:21.0 is given in table below.

S.No.	Applicability Conditions	Justification
1.	This methodology is applicable to grid-connected renewable energy power generation project activities that: (a) Install a Greenfield power plant; (b) Involve a capacity addition to (an) existing plant(s); (c) Involve a retrofit of (an) existing operating plants/units; (d) Involve a rehabilitation of (an) existing plant(s)/unit(s); or (e) Involve a replacement of (an) existing plant(s)/unit(s).	The project activity includes an installation of a grid-connected wind power plant of 160MW capacity at the project site where no renewable energy power plant was operated prior to the implementation of the project activity. Therefore, the project is a Greenfield project.
2.	In case the project activity involves the integration of a BESS, the methodology is applicable to grid-connected renewable energy power generation project activities that:	The project involves an installation of a Greenfield wind power plant and will not install BESS..

<table>
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<td>(a) Integrate BESS with a Greenfield power plant;
(b) Integrate a BESS together with implementing a capacity addition to (an) existing solar photovoltaic1 or wind power plant(s)/unit(s);
(c) Integrate a BESS to (an) existing solar photovoltaic or wind power plant(s)/unit(s) without implementing any other changes to the existing plant(s);
(d) Integrate a BESS together with implementing a retrofit of (an) existing solar photovoltaic or wind power plant(s)/unit(s).

Table 2. Combinations of renewable energy technologies and mode of BESS applicable for integration
<table>
<tr><td>Renewable Energy Technology Mode of installation of BESS</td><td>Solar photovoltaic or wind</td><td>Other renewable technologies</td></tr>
<tr><td>BESS + (a) Greenfield plant(s)</td><td>Eligible</td><td>Eligible</td></tr>
<tr><td>-BESS+ capacity addition to existing plant(s)</td><td>Eligible</td><td>Not eligible</td></tr>
<tr><td>BESS with no other changes to the existing plant(s)</td><td>Eligible</td><td>Not eligible</td></tr>
<tr><td>BESS + retrofit to existing plant(s)</td><td>Eligible</td><td>Not eligible</td></tr>
</table>
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<tr>
<td rowspan="2">3.</td>
<td>The methodology is applicable under the following conditions:
(a) Hydro power plant/unit with or without reservoir, wind power plant/unit, geothermal power plant/unit, solar power plant/unit, wave power plant/unit or tidal power plant/unit;</td>
<td>The project involves an installation of a Greenfield wind power plant and is eligible under the applied methodology.</td>
</tr>
<tr>
<td>(b) In the case of capacity additions, retrofits, rehabilitations or replacements (except for wind, solar, wave or tidal power capacity addition</td>
<td>The condition is not applicable as the project involves an installation of a Greenfield wind power plant.</td>
</tr>
</table>

	projects) the existing plant/unit started commercial operation prior to the start of a minimum historical reference period of five years, used for the calculation of baseline emissions and defined in the baseline emission section, and no capacity expansion, retrofit, or rehabilitation of the plant/unit has been undertaken between the start of this minimum historical reference period and the implementation of the project activity.	
	(c) In case of Greenfield project activities applicable under paragraph 5 (a) above, the project participants shall demonstrate that the BESS was an integral part of the design of the renewable energy project activity (e.g. by referring to feasibility studies or investment decision documents);	The condition is not applicable as the project involves an installation of a Greenfield wind power plant and will not install BESS.
	(d) The BESS should be charged with electricity generated from the associated renewable energy power plant(s). Only during exigencies 2 may the BESS be charged with electricity from the grid or a fossil fuel electricity generator. In such cases, the corresponding GHG emissions shall be accounted for as project emissions following the requirements under section 5.4.4 below. The charging using the grid or using fossil fuel electricity generator should not amount to more than 2 per cent of the electricity generated by the project renewable energy plant during a monitoring period. During the time periods (e.g. week(s), months(s)) when the BESS consumes more than 2 per cent of the electricity for charging, the project participant shall not be entitled to issuance of the certified emission reductions for the concerned periods of the monitoring period.	The condition is not applicable as the project involves an installation of a Greenfield wind power plant and will not install BESS.
4.	In case of hydro power plants, one of the following conditions shall apply: (a) The project activity is implemented in existing single or multiple reservoirs, with no change in the volume of any of the reservoirs; or	The condition is not applicable as the project is not a hydro-power plant.
	(b) The project activity is implemented in existing single or multiple reservoirs, where the volume of the reservoir(s) is increased and the power density,	

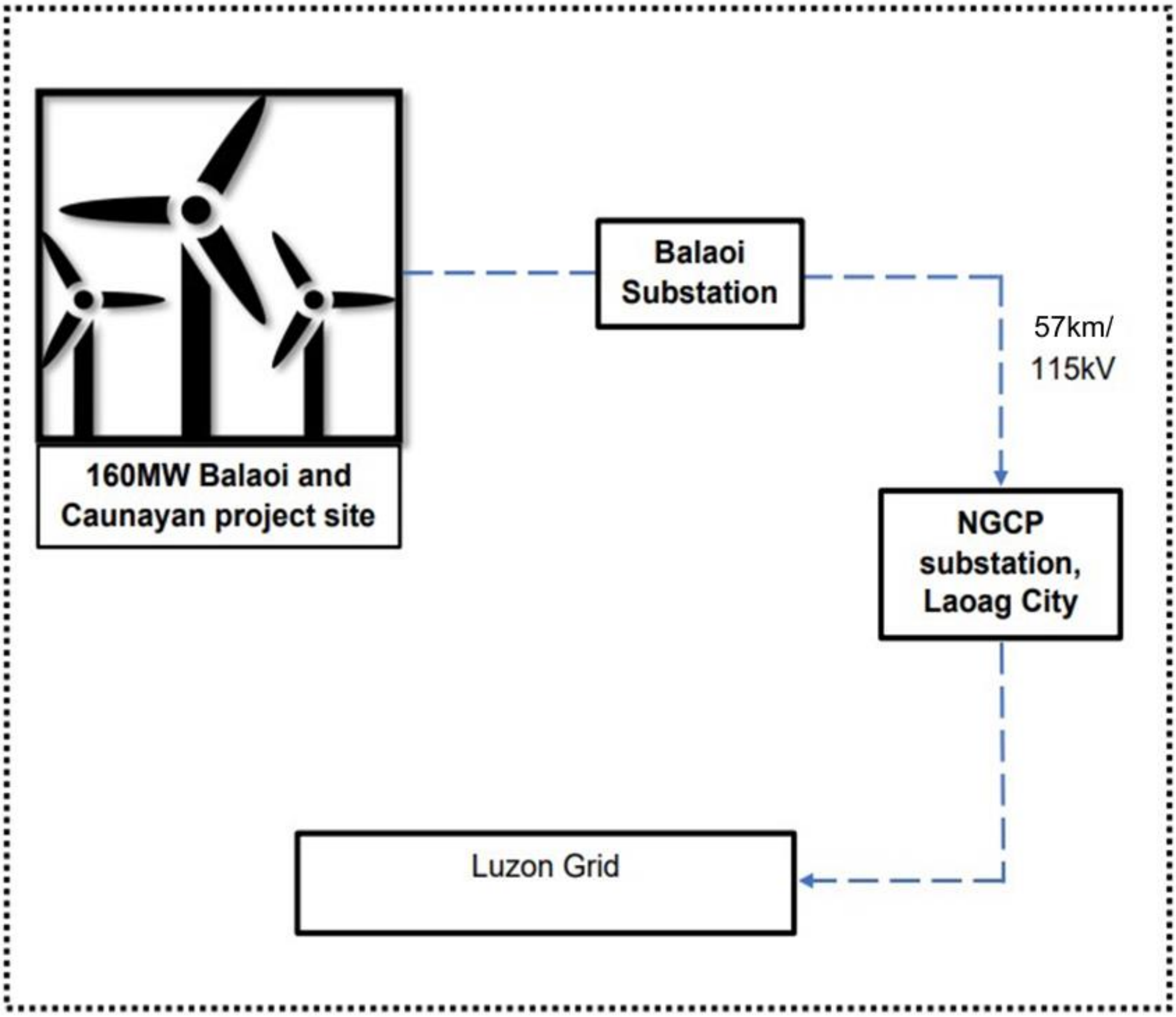
	calculated using equation (7), is greater than 4 W/m2; or	
	(c) The project activity results in new single or multiple reservoirs and the power density, calculated using equation (7), is greater than 4 W/m2; or	
	(d) The project activity is an integrated hydro power project involving multiple reservoirs, where the power density for any of the reservoirs, calculated using equation (7), is lower than or equal to 4 W/m2, all of the following conditions shall apply: i. The power density calculated using the total installed capacity of the integrated project, as per equation (8), is greater than 4 W/m2; ii. Water flow between reservoirs is not used by any other hydropower unit which is not a part of the project activity; iii. Installed capacity of the power plant(s) with power density lower than or equal to 4 W/m2 shall be: a) Lower than or equal to 15 MW; and b) Less than 10 per cent of the total installed capacity of integrated hydro power project.	
5.	In the case of integrated hydro power projects, project proponent shall: (a) Demonstrate that water flow from upstream power plants/units spill directly to the downstream reservoir and that collectively constitute to the generation capacity of the integrated hydro power project; or	The condition is not applicable as the project is a wind power project and not an integrated hydro power project.
	b) Provide an analysis of the water balance covering the water fed to power units, with all possible combinations of reservoirs and without the construction of reservoirs. The purpose of water	

	balance is to demonstrate the requirement of specific combination of reservoirs constructed under CDM project activity for the optimization of power output. This demonstration has to be carried out in the specific scenario of water availability in different seasons to optimize the water flow at the inlet of power units. Therefore, this water balance will take into account seasonal flows from river, tributaries (if any), and rainfall for minimum of five years prior to the implementation of the CDM project activity.	
6.	The methodology is not applicable to: (a) Project activities that involve switching from fossil fuels to renewable energy sources at the site of the project activity, since in this case the baseline may be the continued use of fossil fuels at the site; (b) Biomass fired power plants/units.	The condition is not applicable as: a) The project activity involves an installation of a Greenfield wind power plant. b) The project is not a Biomass fired power plant.
7.	In the case of retrofits, rehabilitations, replacements, or capacity additions, this methodology is only applicable if the most plausible baseline scenario, as a result of the identification of baseline scenario, is "the continuation of the current situation, that is to use the power generation equipment that was already in use prior to the implementation of the project activity and undertaking business as usual maintenance".	The condition is not applicable as the project involves an installation of a Greenfield wind power plant.

B.3. Project boundary

With reference to ACM0002, Version:21.0, "The spatial extent of the project boundary includes the project power plant/unit and all power plants/units connected physically to the electricity system that the CDM project power plant is connected to." The project boundary incorporates 160MW Balaoi & Caunayan Wind Energy Project, substation, all power plants connected physically to the national grid to which the proposed project will also be connected. Diagram showcasing the project boundary is given below.

Source		GHGs	Included?	Justification/Explanation
Baseline scenario	CO2 emissions from electricity generation in fossil fuel fired power plants	CO_2	Yes	Main Emission Source
		CH_4	No	Minor Emission Source
		N_2O	No	Minor Emission Source
Project scenario	Greenfield Wind Energy Project	CO_2	No	As per ACM002, a wind energy project produces zero emissions.
		CH_4	No	
		N_2O	No	



Flow Diagram of Project Boundary

B.4. Establishment and description of baseline

As per ACM0002, Version:21.0 methodology, the baseline scenario for a project involving an installation of a Greenfield grid-connected renewable energy project should be as following:

"If the project activity is the installation of a Greenfield power plant, the baseline scenario is electricity delivered to the grid by the project activity would have otherwise been generated by the operation of grid-connected power plants and by the addition of new generation sources, as reflected in the combined margin (CM) calculations described in "TOOL07: Tool to calculate the emission factor for an electricity system".
The proposed project involves construction and operation of 160MW Wind Power plant which will harness wind energy to generate electricity and supply to the grid. Primarily, the Philippines's grid is powered by fossil-fuel based power plants. Hence, in the baseline scenario, equal amount of electricity to be supplied by the fossil-fuel based power plants connected to the grid.

To calculate the baseline emissions from the project, the combined margin emission factor of the national grid (EFgrid,CM,y) will be used and it is calculated according to Tool 07 - "Tool to calculate the emission factor for an electricity system", version 07.0. Data published by the Department of Energy of the host country, Philippines[2], on their official website will be used for Combined Margin (CM) Emission Factor for wind and solar power projects. The emission factor calculation will be detailed further in section B.6 given below.

B.5. Demonstration of additionality

According to version 21.0 of ACM0002, the demonstration of additionality for the proposed Project activity is being carried out in accordance with the additionality tool provided by the UNFCCC i.e. "Tool for demonstration and assessment of Additionality" Version 07.0.0.
The tool provides a step-wise approach to demonstrate additionality, which is displayed below:

<u>Step 0: Demonstration whether the proposed project activity is the first-of its-kind</u>

This step is not applied to the project activity since it is not first-of-its-kind, hence the additionality of the project will be demonstrated in next steps below.

<u>Step 1: Identification of alternatives to the project activity consistent with current laws and regulations</u>

Sub-step 1a: Define alternatives to the project activity:

[2] https://www.doe.gov.ph/electric-power/2015-2017-national-grid-emission-factor-ngef

Paragraph 8 of version 07.0.0 of the additionality tool states: "Project activities that apply this tool in context of approved consolidated methodology ACM0002, only need to identify that there is at least one credible and feasible alternative that would be more attractive than the proposed project activity."

We will therefore consider the two scenarios in the following analysis:

Alternative 1: The proposed project undertaken without the GS registration.

Alternative 2: Continuation of the current situation. In this case, the proposed project will not be constructed and the power will be solely supplied from the national grid.

Sub-step 1b: Consistency with mandatory laws and regulations:

The alternative 2 "the continuation of the current situation" does not face with any barrier from the current law and regulation because it is the "do-nothing" alternative. The project owner of a proposed project has no obligation to build or invest in the power plant to supply electricity. In this scenario, the equal amount of electricity will be supplied from the power plants running on coal as it is the largest and primary source of electricity in the country. Hence this alternative is consistent with mandatory laws and regulations.

The alternative 1 is consistent with mandatory laws and regulations (such as RE Act, 2008, Philippine Energy Plan 2020-2040 that emphasize on development, utilization and commercialization of RE resources in the country and increase RE supply in the total energy mix. of the country.) as it has received all the necessary approvals and permits from the national and local authorities.

Both the alternatives enlisted above are found to comply with the mandatory laws and regulations taking into account the enforcement of the legislations in the region or country and EB decisions on national and/or sectoral policies and regulations.

However, alternative 2 has been selected as the appropriate baseline alternative for this project activity in line with the methodology.

Step 2: Investment analysis

Determine whether the proposed project activity is not:

(a) The most economically or financially attractive; or

(b) Economically or financially feasible, without the revenue from the sale of emission reductions

To conduct the investment analysis, the following sub-steps are applied:

Sub-step 2a: Determine appropriate analysis method

The "Tool for the demonstration and assessment of additionality", Version 7.0.0 provides three methods of investment analysis: simple cost analysis (Option I), investment comparison analysis (Option II) and benchmark analysis (Option III).

The proposed project activity generates financial and economic benefits from sale of the generated electricity, so the simple cost analysis (Option I) is not applicable. Out of the two remaining options, Option II is also not applicable as there are no other credible and realistic baseline scenario alternatives other than electricity supply from the grid. Thus, the benchmark analysis (Option III) is chosen to prove additionality.

Sub-step 2b (Option III): Apply benchmark analysis

The investment analysis using Benchmark analysis approach (Option III) has been chosen.

In the following, Equity IRR is used to demonstrate the additionality of the project.

Selection of Appropriate Benchmark

The benchmark has been considered in accordance with paragraph 16 of the Investment analysis tool 27 V 10.0, "In situations where an investment analysis is carried out in nominal terms and the available IRR benchmarks are in real terms, project participants shall convert the real term values of benchmarks to nominal values by adding the inflation rate."

Methodology deployed for arriving at a suitable value of Benchmark using Default Value has been described below:

- As the proposed project activity generates power utilizing wind energy, Group 1 as per para 5a of Investment analysis tool, version 10.0 which was applicable at the time of investment decision, has been identified as a suitable category.
- The investment analysis has been carried out in Nominal terms. Accordingly, Default value as given in table under the Appendix, of Investment Analysis tool, version 10.0 has been adjusted by adding suitable forecasted inflation rate taken from IMF (2023 to 2027).

The benchmark applicable at the time of investment decision time has been computed in the following manner:

Default Value Benchmark:

The cost of equity is determined by selecting the values provided in the table of the Appendix, i.e. Default values for cost of equity (expected return on equity) in the 'Methodological tool: Investment analysis' Version 10.0.

The Required return on equity (benchmark) was computed in the following manner:

Nominal Benchmark = {(1+Real Benchmark)*(1+Inflation rate)}-1

Where:

Default value for Real Benchmark = 10.24% (as per Investment Analysis Tool, version 10.0) and Inflation Rate forecast for by IMF (2023 to 2027).

Default Value as per UNFCCC guidelines	10.24%	Default value for Philippines as per CDM Tool 27 Version 10.0
Inflation forecast IMF for 5 years	3.70%	As per IMF (2023-2027)
Default Value Benchmark (with 5yrs Forecast)	14.32%	in nominal terms
Expected Return on Equity as per Default Value	**14.32%**	in nominal terms

From above, a nominal benchmark of 14.32% has been selected as an appropriate benchmark.

- The date of investment decision is 6th Oct, 2020, the date on which NTP was signed.

Sub-step 2c: Calculation and comparison of financial indicators

The key assumptions used to calculate the Project IRR of the proposed project are presented in the table below:

Key assumptions to calculate the Equity IRR

No	Parameter	Unit	Value	Source
1	Installed capacity	MW	160	Feasibility Study Report, Service Maintenance and Availability Agreement
2	Wind turbine model		Siemens Gamesa	Schedule G4
3	Number of turbines		32	Feasibility Study Report, Service Maintenance and Availability Agreement
4	Turbine capacity	MW	5	Feasibility Study Report, Service Maintenance and Availability Agreement
5	Expected COD		15-Dec-22	Estimated
7	Net Electricity(P75) Supplied to the	MWh	532,300	Energy Yield Assessment Report and BWPC Wind

	Grid Considering Line Losses per year			Farm Consumption Report
8	TOTAL PRIMARY COSTS	Peso ('000)	8,615,143	Based on BWPC's development experience
9	TOTAL OTHER COSTS	Peso ('000)	1,903,147	Based on BWPC's development experience
10	TOTAL FINANCING COSTS	Peso ('000)	93,342	Based on BWPC's development experience
11	Contingency	Peso ('000)	795,875	Based on BWPC's development experience
12	Total investment cost	Peso ('000)	11,407,507	Project Financial Model
13	Period of financial assessment	year	20	Schedule G4
14	Electricity Tariff (WESM)	Peso/kWh	3.65	Project Feasibility Report
15	Line rental fee	Peso/kWh	0.30	Project Financial Model
16	Net Tariff (WESM)	Peso/kWh	3.35	Calculated
17	WTG Maintenance	Peso ('000)	72,096	Project Financial Model
18	Personal Expenses	Peso ('000)	47,000	Project Financial Model
19	Professional Fees (O&M Consultant, Legal, Wind analysis, Accounting, SAP)	Peso ('000)	10,400	Project Financial Model, Philippines Wind Operating Expenses
20	Salvage value	Peso	0	As per Tool 27 Version 12.0
21	Depreciation	100%	Straight Line	Project Financial Model
22	**Insurance ('000)**			Project Financial Model
23	All risks		0.50%	of primary costs
24	3rd Party liability, PHP	Peso ('000)	1500	Project Financial Model
25	Business Interruption		0.70%	of Gross Profit
26	**Real Property Tax ('000)**			Project Financial Model
27	Real Property Tax Rate		1.5%	https://cta.judiciary.gov.ph/home/download/eab2f1210348a90ab253b2d765902762 Section 15, Page 25
28	RPT Basis as % of Depreciated Primary Costs		75%	https://www.bir.gov.ph/index.php/tax-code.html SEC. 34.(F)

				https://www.bir.gov.ph/index.php/tax-information/capital-gains-tax.html FAQ No.2, Point 3
29	Post-Tax Equity IRR	%	8.89%	Calculated

The above estimates show a Post-Tax Equity IRR of 8.89% which is lower than the chosen benchmark of 14.32%.

Sub-step 2d: Sensitivity Analysis

A sensitivity analysis of the project activity has been conducted to test the robustness of the above calculations. Under the guidance in Tool27 – "Investment analysis" – Version 10.0, paragraph 27, which states that variables, including the initial investment cost, that constitute more that 20% of either total project costs or total project revenues should be subjected to reasonable variations.

The following parameters are used in the sensitivity analysis of the project activity:

- Total investment cost
- Electricity Output
- Electricity Tariff
- O&M Cost

Table below shows the impact of variations in key factors on the Post-Tax Equity IRR considering a ±10% variation in the parameters.

Sensitivity analysis

Variation %	-10%	0	+10%	Variation required to breach benchmark	Value required to breach benchmark
Investment Cost	10.27%	8.89%	7.69%	-32%	7,757,105
Electricity Output	7.30%	8.89%	10.39%	39%	739,897
Electricity Tariff	7.14%	8.89%	10.53%	36%	4.964
O&M Costs	8.96%	8.89%	8.82%	-819%	-518370.24

An analysis has been done to identify the percentage variation at which the financial indicators will equal/breach the benchmark and the probability of its occurrence. Based on sensitivity analysis it can be concluded that the proposed project activity is additional with reasonable variation in values and is not likely to reach the benchmark value. The occurrence of these events is unlikely for the following reasons:

a) **Investment Cost:** The project cost considered for investment analysis is 11,407,507 Peso ('000). A variation of -32% is required for IRR to breach benchmark is not possible since the project is going to be commissioned and is finalized it is unlikely that the project cost will be decreased below the breaching value. The investment cost per MW for the Project Activity is 71,297 Peso ('000). Comparing to the investment cost per MW of 2 similar projects from the host country, "81MW Caparispisan Wind Energy Project" and "52MW Bangui Bay Wind Farm" owned by the same sponsor, ACEN Corporation, the investment cost of the Project Activity needs to be 53.74% higher (109,613 Peso ('000)) and 23.57% higher (88,104 Peso ('000)) respectively, which is highly unlikely.
b) **Electricity Output:** The project electricity output is 532,300 MWh. Given the analysis done in the Energy Yield Assessment Report and BWPC Wind Farm Consumption Report, it is unlikely the project will reach 39% of the electricity output and cross the benchmark.
c) **Electricity Tariff:** The tariff rate considered at the time of the investment decision is 3.65 Peso/kWh. A variation of 36% is required to breach the IRR benchmark. Since the tariff rate considered at the time of investment decision is not changing, it is unlikely the IRR will breach the benchmark. The Project Developer was awarded a tariff of 4.4386 Peso/kWh through the Philippine Green Energy Auction Program (GEAP) through bidding 2 years after the investment decision. The fixed-rate tariff of 4.4386 Peso/kWh needs to be 11.84% higher than the ROE Benchmark to breach the tariff value of 4.964 Peso/kWh. The fixed-rate tariff is fixed for the lifetime of Project Activity, i.e., 20 years. So it is not possible for the tariff to breach the benchmark.
d) **O&M Costs:** The sensitivity analysis reveals that O&M will breach the benchmark at negative values and is hypothetical case. Since the O&M cost is subject to escalation (as evidence by the O&M agreement) and also subject to inflationary pressure, any reduction in the O&M costs is highly unlikely. The O&M Costs per MW for the Project Activity is 2,636 Peso ('000). Comparing to the investment cost per MW of 2 similar projects from the host country, "81MW Caparispisan Wind Energy Project" and "52MW Bangui Bay Wind Farm" owned by the same sponsor, ACEN Corporation, the O&M Costs of the Project Activity needs to be 92.83% higher (5,083 Peso ('000)) and 81.41% higher (4,782 Peso ('000)) respectively, which is highly unlikely.

Hence, the sensitivity analysis shows that there is unlikely to be a scenario in which the variation of a parameter can make the project IRR reach the benchmark.

In conclusion, the proposed project activity is unlikely to be financially attractive.

<u>Step 3: Barrier analysis</u>

Barrier analysis has not been applied.

<u>Step 4: Common practice analysis</u>

Sub-step 4a: Analyze other activities similar to the proposed project activity

Stepwise approach for common practice analysis has been carried out as per Methodological Tool24 - "Common Practice", Version 03.1.

<u>Step (1): Calculate applicable capacity or output range as +/-50% of the total design capacity or output of the proposed project activity.</u>

The proposed project activity has the installed capacity of 160 MW. So applicable capacity range as +/-50% is from 80 MW to 240 MW.

<u>Step (2): Identify similar projects (both CDM and non-CDM) which fulfil all of the following conditions:</u>

a) The projects are located in the applicable geographical area;
b) The projects apply the same measure as the proposed project activity;
c) The projects use the same energy source/fuel and feedstock as the proposed project activity, if a technology switch measure is implemented by the proposed project activity;
d) The plants in which the projects are implemented produce goods or services with comparable quality, properties and applications areas (e.g. clinker) as the proposed project plant;
e) The capacity or output of the projects is within the applicable capacity or output range calculated in Step 1;
f) The projects started commercial operation before the project design document (CDM-PDD) is published for global stakeholder consultation or before the start date of proposed project activity whichever is earlier for the proposed project activity.

Identification of the similar projects (CDM and non-CDM) is carried out as per sub-steps of Step (2) as follows:

a) The proposed project is located in the Philippines so the entire host country (Philippines) was chosen as the applicable geographical area.
b) The proposed project activity is a Greenfield wind power plant. Therefore, all the wind power plants (the same measure as the proposed project activity) are candidates for similar projects.
c) The energy source used by the proposed project activity is wind. Hence, only wind power plants are candidates for similar projects.
d) The proposed project activity produces electricity from wind energy. Therefore, all the wind power plants connected to the national grid are candidates for similar projects.

e) The capacity range of the similar projects is within the applicable capacity range from 80 MW to 240 MW.
f) The start date of the project under the Methodological tool for common practice is the date on which the project participants commit to making expenditures for the construction or modification of the main equipment or facility (e.g. a wind turbine), or for the provision or modification of a service (e.g. distribution of energy-efficient light bulbs, change of transport management system). Where a contract is signed for such expenditures (e.g. for procurement of a wind turbine), it is the date on which the contract is signed. In other cases, it is the date on which such expenditures are incurred. If the project activity involves more than one of such contracts or incurred expenditures, it is the first of the respective dates. Activities incurring minor pre-project expenses (e.g. feasibility studies, preliminary surveys) are not considered in the determination of the start date.

The Contracts for the Turbine Supply Agreement and Installation and Commissioning Agreement between Bayog Wind Power Corp. Company, Siemens Gamesa Renewable Energy Technology (China) Co., Ltd Company and Gamesa Eolica, S.L. Unipersonal – Philippine Branch Company are the first contracts for expenditures of the proposed project activity, signed on 14 May, 2021 and thus, it is considered as the start date of the project activity for common practice analysis. The wind power projects in the Philippines, which have started commercial operation before 14/05/2021 have been considered for analysis. These projects are listed as below:

List of wind power projects considered in the Philippines

No.	Name	Capacity	Type	Commissioning year	Under carbon-credit scheme	Investor
1	Burgos Wind Project	150	On-shore	11/11/2014	CDM	EDC Burgos Wind Power Corporation
2	81MW Caparispisan Wind Energy Project	81	On-shore	09/11/2014	GS	North Luzon Renewable Energy Corp.

There are only two projects that are within the applicable capacity range from 80 MW to 240 MW. Thus, numbers of similar projects identified are 2.

Step (3): Within the projects identified in Step (2), identify those that are neither registered CDM project activity, project activity submitted for registration, nor project activity undergoing validation. Note their number N_{all}.

As indicated above, both are registered projects, therefore, $\mathbf{N_{all} = 0}$.

Step (4): Within similar projects identified in Step 3, identify those that apply technologies that are different to the technology applied in the proposed project activity. Note their number N_{diff}.

As $\mathbf{N_{all} = 0}$ so, $\mathbf{N_{diff} = 0}$

Step (5): Calculate factor $F=1-N_{diff}/N_{all}$ representing the share of similar projects (penetration rate of the measure/technology) using a measure/technology similar to the measure/technology used in the proposed project activity that deliver the same output or capacity as the proposed project activity.

Calculate factor: $F = 1 - N_{diff}/N_{all} = 1 - 0/0 = 1$

$N_{all} - N_{diff} = 0$

As per methodological tool "Common practice" - Version 03.1, the proposed project activity is a "common practice" within a sector in the applicable geographical area if the factor F is greater than 0.2 and N_{all} - N_{diff} is greater than 3.

The above conditions are not fulfilled, F = 1 is greater than 0.2 and $N_{all} - N_{diff} = 0$ is smaller than 3. Therefore, the project activity is not a "common practice" within the sector in the applicable geographical area, according to the guideline.

In conclusion, the proposed project activity is additional.

B.5.1 Prior Consideration

In accordance with GS4GG's Principles and Requirements, Version:1.2, "retroactive projects shall submit the required documents for preliminary review (time of first submission) within one year of the project start date." The start date of the project activity is 14/05/2021 and the project had submitted the required documents for preliminary review submission within one year of the project start date. Thus, the project fulfills Gold Standard Certification eligibility criteria for retroactive projects as well as clearly demonstrates prior consideration of revenues from Gold Standard certification.

B.5.2 Ongoing Financial Need

Not applicable

B.6. Sustainable Development Goals (SDG) outcomes

Relevant Target/Indicator for each of the three SDGs

Sustainable Development Goals Targeted	Most relevant SDG Target	SDG Impact
		Indicator (Proposed or SDG Indicator)
13 Climate Action (mandatory)	13.2 Integrate climate change measures into national policies, strategies and planning	13.2.1 Number of countries with nationally determined contributions, long-term strategies, national adaptation plans, strategies as reported in adaptation

		communications and national communications. Target: 363,894 tCO_2e/annum
7 Affordable and Clean Energy	7.2 By 2030, increase substantially the share of renewable energy in the global energy mix	7.2.1 Renewable energy share in the total final energy consumption. Target: 532,300 MWh/year
8 Decent Work and Economic Growth	8.5 By 2030, achieve full and productive employment and decent work for all women and men, including for young people and persons with disabilities and equal pay for work of equal value	8.5.1 Average hourly earnings of employees, by sex, age, occupation and persons with disabilities Target: To provide employment to around 10 local people including women with salaries for everyone equal or higher than minimum wage as prescribed by the Labour Code of the country. Equal opportunities in employment will be available for all, regardless of discrimination on any basis.

B.6.1 Explanation of methodological choices/approaches for estimating the SDG Impact

SDG 7: Ensure access to affordable, reliable, sustainable and modern energy for all
This SDG outcome is measured in principle with ACM0002 “Grid-connected electricity generation from renewable sources”, Version:21.0 for the quantity of net electricity transmitted to the grid by the project (EGfacility,y).

Baseline:

The Philippines is highly dependent upon fossil-fuel based power plants to fulfill its ever-increasing energy consumption. Hence, in the baseline scenario, the national grid is fed with electricity generated through fossil-fuel sources.

Project Outcome:

The project will annually generate a net electricity of 532,300 MWh/year of clean, renewable and sustainable electricity to the national electricity grid. Hence, it will contribute towards the Target 7.2: By 2030, increase substantially the share of renewable energy in the global energy mix.

Where:

The Philippine's grid is powered by fossil-fuel based power plants. Hence, in the baseline scenario, clean, renewable and sustainable electricity produced and supplied to the grid = 0 (Baseline Estimate)
In the Project Scenario, clean, renewable and sustainable electricity to be produced and supplied by the project to the grid = 532,300 MWh/yr (Project Estimate)
Therefore, net clean, renewable and sustainable electricity that will be produced and supplied to the grid = 532,300 MWh/yr (Net Benefit)

SDG 8: Promote sustained, inclusive and sustainable economic growth, full and productive employment and decent work for all.
Baseline:
In the case of baseline scenario, no such employment opportunities would have been available for the local communities.
Project Outcome:
The project will benefit the local communities as local residents will be prioritized for employment opportunities, based on their qualifications and skills. Around 10 people including women will be provided with employment opportunities. Being an equal opportunity company, BWPC will have equal work opportunities open to all, irrespective of their origin, race, religion, caste, gender, language, disability or any other status. Moreover, salaries of all the employees will be equal or higher than the minimum wage as per the law. Hence, the project will bring in economic stability, alleviate poverty, improve standard of living and will eventually help in improving the regional economy. Also, in order to develop skills and train employees in various aspects, all the employees will be given induction trainings regarding technical skill development and health and safety so as to provide qualitative employment and safe and healthy work environment. Therefore, the project contributes to the Target 8.5: By 2030, achieve full and productive employment and decent work for all women and men, including for young people and persons with disabilities, and equal pay for work of equal value.
Where:
In the baseline scenario, no. of jobs offered to locals = 0 (Baseline Estimate)
In the project scenario, no. of jobs to be offered to locals = 10 (Approx., Project Estimate)
Therefore, jobs to be provided to locals = Approx. 10 (Net Benefit)

SDG 13: Take urgent action to combat climate change and its impacts
This project activity will reduce 363,894 tons of CO2 equivalent/year and thus, the project will contribute to the National SDG 13 - Target 13.2: Integrate climate change measures into national policies, strategies and planning. This SDG outcome is measured in accordance with ACM0002 "Grid-connected electricity generation from renewable sources", Version:21.0 for the emission reductions generated by the project, ERy (tCO2e).
Baseline Emissions (BEy)

Baseline emissions include only CO_2 emissions from electricity generation in fossil fuel fired power plants that are displaced due to the project activity. It is calculated as shown below:

$$BE_y = EG_{PJ,y} \times EF_{grid,CM,y}$$

Where:

BE_y = Baseline emissions in year y (tCO_2e/yr)

$EG_{PJ,y}$ = Quantity of net electricity generation that is produced and fed into the grid as a result of the implementation of the CDM project activity in year y (MWh/yr)

$EF_{grid,CM,y}$ = Combined margin CO_2 emission factor for grid connected power generation in year y calculated using the latest version of "TOOL07: Tool to calculate the emission factor for an electricity system" (tCO_2e/MWh)

Calculation of EGPJ,y

The project activity is an installation of new grid-connected renewable energy power plant at a site where no renewable energy power plant was operated prior to the implementation of the project activity, then:

$$EG_{PJ,y} = EG_{facility,y}$$

Where:

$EG_{PJ,y}$ = Quantity of net electricity generation that is produced and fed into the grid as a result of the implementation of the CDM project activity in year y (MWh/yr)

$EG_{facility,y}$ = Quantity of net electricity generation supplied by the project plant/unit to the grid in year y (MWh/yr)

Therefore, the baseline emissions are to be calculated as follows:

$$BE_y = EG_{facility,y} \times EF_{grid,CM,y}$$

In order to calculate Combined Margin CO_2 emission factor ($EF_{grid,CM,y}$), the methodology refers to "Tool to calculate the emission factor for an electricity system". The Combined margin has already been published by Department of Energy of Philippines on their official website[3].

Baseline emission factor is calculated as combined margin, a combination of operating margin and build margin factors according to the procedure given in the "Tool for calculating the emission factor for an electricity system", Version:07.0. The combined margin emission factor is calculated using the following steps:

(a) Step 1: Identify the relevant electricity systems.
(b) Step 2: Choose whether to include off-grid power plants in the project electricity system (optional).
(c) Step 3: Select a method to determine the operating margin (OM).
(d) Step 4: Calculate the operating margin emission factor according to the selected Method.
(e) Step 5: Calculate the build margin (BM) emission factor.

[3] https://www.doe.gov.ph/electric-power/2015-2017-national-grid-emission-factor-ngef

(f) Step 6: Calculate the combined margin (CM) emission factor.

Also, as per the "Tool to calculate the emission factor for an electricity system", Version:07.0, wind power generation project activities should use wOM = 0.75 and wBM = 0.25 for the first crediting period and for subsequent crediting periods.

$$EFgrid,CM,y = EFgrid,OM,y \times wOM + EFgrid,BM,y \times wBM$$

$$= 0.7122 \times 0.75 + 0.5979 \times 0.25$$
$$= 0.6836\ tCO2e/MWh\ \text{(DOE of Philippines)}$$ [4]

Project Emissions (PEy)

In accordance with methodology ACM0002, Version:21.0, the project emissions are calculated using the equation given below:

$$PEy = PEFF,y + PEGP,y + PEHP,y$$

Where:

PEy = Project emissions in year y (tCO2e/yr)
PEEF,y = Project emissions from fossil fuel consumption in year y (tCO2e/yr)
PEGP,y = Project emissions from the operation of dry, flash steam or binary geothermal power plants in year y (tCO2e/yr)
PEHP,y = Project emissions from water reservoirs of hydro power plants in year y (tCO2e/yr)

The proposed project is a wind energy project and therefore, PEy=0.

Leakage

As per the methodology ACM0002, Version:21.0, no other leakage emissions are considered. In case of electric sector projects, the emissions potentially arising due to activities such as power plant construction and upstream emissions from fossil fuel use (e.g. extraction, processing, transport etc.) are neglected.

Emission Reductions *ERy*

Emission reductions are calculated as follows:

$$ERy = BEy - PEy$$

Where:

ERy = Emission reductions in year y (tCO2e/yr)
BEy = Baseline emissions in year y (tCO2e/yr)
PEy = Project emissions in year y (tCO2e/yr)

B.6.2 Data and parameters fixed ex ante

SDG13

Data/parameter	EFgrid,OM,y
Unit	tCO2e/MWh

[4] https://www.doe.gov.ph/electric-power/2015-2017-national-grid-emission-factor-ngef

Description	Operating margin CO2 emission factor for grid connected power generation in year y
Source of data	2015-2017 National Grid Emission Factor (NGEF) published by DOE of Philippines
Value(s) applied	0.7122
Choice of data or Measurement methods and procedures	The EFgrid,OM,y has been calculated by the DOE of Philippines.
Purpose of data	To calculate CO2 emission reductions.
Additional comment	Operating Margin will remain fixed throughout the crediting period.

Data/parameter	EFgrid,BM,y
Unit	tCO2e/MWh
Description	Build margin CO2 emission factor for grid connected power generation in year y
Source of data	2015-2017 National Grid Emission Factor (NGEF) published by DOE of Philippines
Value(s) applied	0.5979
Choice of data or Measurement methods and procedures	The EFgrid,BM,y has been calculated by the DOE of Philippines.
Purpose of data	To calculate CO2 emission reductions.
Additional comment	Build Margin will remain fixed throughout the crediting period.

Data/parameter	EFgrid,CM,y
Unit	tCO2e/MWh
Description	Combined margin CO2 emission factor for grid connected power generation in year y
Source of data	2015-2017 National Grid Emission Factor (NGEF) published by DOE of Philippines
Value(s) applied	0.6836
Choice of data or Measurement methods and procedures	The EFgrid,CM,y is calculated using Operating margin CO2 emission factor - EFgrid,OM,y and Build margin CO2 emission factor - EFgrid,BM,y and has been calculated by the DOE of Philippines.

Purpose of data	To calculate CO2 emission reductions.
Additional comment	Combined Margin will remain fixed throughout the crediting period.

B.6.3 Ex ante estimation of SDG Impact

SDG 7: Affordable and Clean Energy – The project will generate 532,300 MWh/yr.
Where:
The Philippine's grid is powered by fossil-fuel based power plants. Hence, in the baseline scenario, clean, renewable and sustainable electricity produced and supplied to the grid = 0 (Baseline Estimate)
In the Project Scenario, clean, renewable and sustainable electricity to be produced and supplied by the project to the grid = 532,300 MWh/yr (Project Estimate)
Therefore, net clean, renewable and sustainable electricity that will be produced and supplied to the grid = 532,300 MWh/yr (Net Benefit)

SDG 8: Decent Work and Economic Growth – Employment opportunities will be provided to about 10 local people including women with salaries for everyone equal or higher than minimum wage as prescribed by the Labour Code of the country. Equal opportunities in employment will be available for all, regardless of discrimination on any basis.
Where:
In the baseline scenario, no. of jobs to be offered to locals = 0 (Baseline Estimate)
In the project scenario, no. of jobs to be offered to locals = 10 (Approx., Project Estimate)
Therefore, jobs to be provided to locals = Approx. 10 (Net Benefit)

Ex Ante calculation of SDG13 Impact: Climate Action

Baseline Emissions (BEy)

$$BEy = EGfacility,y \times EFgrid,CM,y$$

Where:
EGfacility,y = 532,300 MWh/year
EFgrid,CM,y (tCO2e/MWh) = 0.6836 tCO2e/MWh

$$BEy = 532{,}300 \times 0.6836$$
$$= 363{,}894 \text{ tCO2e/year}$$

Project Emissions (PEy)

In accordance with ACM0002 methodology, Version:21.0, the project emissions will be zero as the project is a wind energy project.

Leakage

As per the methodology ACM0002, Version:21.0, no other leakage emissions are considered. In case of electric sector projects, the emissions potentially arising due to activities such as power plant construction and upstream emissions from fossil fuel use (e.g. extraction, processing, transport etc.) are neglected.

Emission Reductions (ERy)

Emission reductions are calculated as follows:

$$ERy = BEy - PEy$$

Where:

ERy (Emission reductions in year y) (tCO_2e/yr)

BEy (Baseline emissions in year y) (tCO_2e/yr) = 363,894 $tCO_2e/year$

PEy (Project emissions in year y) (tCO_2e/yr) = 0

Therefore, *ERy* =363,894 tCO_2e/yr

B.6.4 Summary of ex ante estimates of each SDG Impact

SDG 7: Affordable and Clean Energy

Year	Baseline estimate	Project estimate	Net benefit
Year 1	0MWh	532,300 MWh	532,300 MWh
Year 2	0MWh	532,300 MWh	532,300 MWh
Year 3	0MWh	532,300 MWh	532,300 MWh
Year 4	0MWh	532,300 MWh	532,300 MWh
Year 5	0MWh	532,300 MWh	532,300 MWh
Total	**0MWh**	**2,661,500 MWh**	**2,661,500 MWh**

Total number of crediting years	5 years		
Annual average over the crediting period	0MWh	532,300 MWh	532,300 MWh

SDG 8: Decent Work and Economic Growth

The project will benefit the local communities as local residents will be prioritized for employment opportunities, based on their qualifications and skills. Around 10 people including women will be provided with employment opportunities. Being an equal opportunity company, BWPC will have equal work opportunities open to all, irrespective of their origin, race, religion, caste, gender, language, disability or any other status. Moreover, salaries of all the employees will be equal or higher than the minimum wage as per the law. Hence, the project will bring in economic stability, alleviate poverty, improve standard of living and will eventually help in improving the regional economy.

Also, in order to develop skills and train employees in various aspects, all the employees will be given induction trainings regarding technical skill development and health and safety so as to provide qualitative employment and safe and healthy work environment. Therefore, the project contributes to the Target 8.5: By 2030, achieve full and productive employment and decent work for all women and men, including for young people and persons with disabilities, and equal pay for work of equal value.

SDG 13: Climate Action

Year	Baseline estimate	Project estimate	Net benefit
Year 1	363,894 tCO2e	0tCO2e	363,894 tCO2e
Year 2	363,894 tCO2e	0tCO2e	363,894 tCO2e
Year 3	363,894 tCO2e	0tCO2e	363,894 tCO2e
Year 4	363,894 tCO2e	0tCO2e	363,894 tCO2e
Year 5	363,894 tCO2e	0tCO2e	363,894 tCO2e
Total	**1,819,467 tCO2e**	**0tCO2e**	**1,819,467 tCO2e**

Total number of crediting years	5 years		
Annual average over the crediting period	363,894 tCO2e	0tCO2e	363,894 tCO2e

B.7. Monitoring plan

B.7.1 Data and parameters to be monitored

SDG 7: Affordable and Clean Energy

Data / Parameter	$EGfacility,y$
Unit	MWh/yr
Description	Quantity of net electricity generation supplied by the project plant/unit to the grid in year y

Source of data	Metering of the net electricity supplied to the grid
Value(s) applied	532,300 MWh/yr
Measurement methods and procedures	Net Annual Electricity Generation: $EG_{export,y} - EG_{import,y}$ The energy meters will be installed at the interconnection point so as to measure the amount of electricity produced and supplied to the grid by the project (export) and the amount of electricity delivered to the project (import). The net electricity output will be the difference between the export and the import and will be calculated monthly by NGCP and PP.
Monitoring frequency	Daily continuous measurement and monthly recording
QA/QC procedures	Can be verified with invoices raised by the project proponent to NGCP. Also, the energy meters will be calibrated as per the relevant standard requirements.
Purpose of data	To calculate baseline emissions
Additional comment	All the collected data will be archived electronically 2 years after the end of crediting period.

SDG 8: Decent Work and Economic Growth

Data / Parameter	Employment generation
Unit	-
Description	The project will provide work opportunities to about 10 people including women. Employment will be subject to job requirements, qualification of the candidate.
Source of data	Employment contracts and HR Employment Record
Value(s) applied	Total no. of jobs to be provided: 10 (approx.)
Measurement methods and procedures	Employment generation will be monitored through: Employment contracts and HR Employment Record
Monitoring frequency	Once in a monitoring period
QA/QC procedures	Employment Contracts and HR Employment Records shall be prepared and maintained in compliance with the Labour Code of the Philippines.
Purpose of data	To track the contribution made by the project to SDG8.
Additional comment	-

SDG 13: Climate Action

Data / Parameter	ERy
Unit	tCO2e/yr
Description	Emission reductions are calculated as the difference between baseline emissions, project emissions and leakage emissions. Since, leakage emissions are neglected and thus, emission reductions= $ERy = BEy - PEy$ Where: *ERy* (Emission reductions in year y) (tCO2e/yr) BEy (Baseline emissions in year y) (tCO2e/yr) PEy (Project emissions in year y) (tCO2e/yr)
Source of data	Emission Reduction Calculation
Value(s) applied	363,894 tCO2e/yr
Measurement methods and procedures	It is calculated as per the ACM0002 methodology, Version:21.0.
Monitoring frequency	During every verification period
QA/QC procedures	Please refer to EGfacility,y
Purpose of data	Reduction in CO2 emissions due to implementation of the project activity.
Additional comment	-

B.7.2 Sampling plan

Not Applicable

B.7.3 Other elements of monitoring plan

Details regarding metering system, monitoring and calibration: The metering system will confirm to the standards and requirements as per the country's industry guidelines. Prior to the commissioning of project, BWPC will enter in a Metering Service Agreement with NGCP. The meter readings will be measured continuously daily and will be monitored together by the respective representatives from both the parties (BWPC and NGCP) every month. Other than this, the calibration of the energy meters will be done as per standard industry requirements.

Roles and Responsibilities of the monitoring group members: After commissioning of the project, the appointed monitoring team will be responsible for measuring, data collection and storing, monitoring, reporting and operation &

maintenance of metering system and other devices. Monitoring team will include qualified and skilled staff members who will be properly trained according to their role and responsibilities so as to ensure proper functioning and maintenance of the power plant.

Details regarding data collection and archiving: Data recording electronic devices will be used to store monitoring data and the collected data will be archived 2 years after the end of crediting period and the proponent will be responsible for managing and storing the monitoring data.

SECTION C. DURATION AND CREDITING PERIOD

C.1. Duration of project

C.1.1 Start date of project

14/05/2021 (The start date of project is the earliest date on which the Project Developer has committed to expenditures related to the implementation of the Project. The Turbine Supply Agreement has the execution date: 14/05/2021 and thus, confirms the start date of the project.

C.1.2 Expected operational lifetime of project

20 years

C.2. Crediting period of project

C.2.1 Start date of crediting period

25/12/2023 (or the actual commissioning date, whichever is earlier) is the start date of the crediting period.

C.2.2 Total length of crediting period

5 years

SECTION D. SUMMARY OF SAFEGUARDING PRINCIPLES AND GENDER SENSITIVE ASSESSMENT

D.1 Safeguarding Principles that will be monitored

A completed Safeguarding Principles Assessment is in Appendix 1, ongoing monitoring is summarized below.

Principles	Mitigation Measures added to the Monitoring Plan
Principle 3 Community Health, Safety and Working Conditions	BWPC has its own Environment, Health and Safety policy. The policy framework consists of rules, guidelines, procedures, practices, responsibilities which the company and employees need to follow and comply with while performing all the operational activities at the workplace. In order to ensure safety of employees and environmental protection, company will give necessary health and safety up-to-date training to all of its employees. BWPC will conduct annual audit to review EHS plan and policy and besides this, the management will conduct audits periodically so as to evaluate and implement appropriate measures.
Principle 6.1 Labour Rights	The employment will be in compliance with the national employment legislations and occupational health and safety laws will be followed in order to maintain the health and safety of employees at the workplace.
Principle 9.5 Hazardous and Non-Hazardous Waste	During operation phase of the project, the waste will be collected, segregated and disposed of as per local guidelines and waste management laws, through an authorized waste handler.

D.2. Assessment that project complies with GS4GG Gender Sensitive requirements

Question 1 - Explain how the project reflects the key issues and requirements of Gender Sensitive design and implementation as outlined in the Gender Policy?	The proposed project is a Gender-Sensitive project and not a Gender-Responsive project. According to Gold Standard's Gender Policy, a Gender-Sensitive project needs to comply with the following requirement: Foundational gender-sensitive requirement - This strengthens Gold Standard's 'do no harm' approach and addresses safeguards to prevent or mitigate adverse impacts on women or men and girls and boys. Such action is mandatory for all projects seeking Gold Standard certification and includes compliance with the gender 'do no harm' safeguards, gender gap analysis and gender sensitive stakeholder consultations". The project will neither promote nor will be involved in any such activities which will lead to discrimination towards female employees. Employment will be given purely on the basis of one's qualifications, skills and merit and will not depend upon their gender. Also, employees will be given equal pay for equal work done and same contractual terms will be applicable on individuals in same employment.

	Hence, BWPC ensures to provide an inclusive environment at the workplace.
Question 2 - Explain how the project aligns with existing country policies, strategies and best practices	The Philippines is a party to following international agreements and conventions: ● Universal Declaration of Human Rights ● Convention on the Elimination of All Forms of Discrimination against Women (CEDAW) ● Equal Remuneration Convention ● Discrimination (Employment and Occupation Convention) At national level, the country has a national plan called "Gender-Responsive Development", a constitutional framework for gender equality and empowerment of women. The project will comply with all the relevant international and national legislations and will neither be involved nor will promote any activities which will lead to gender inequality and discrimination towards women in any form.
Question 3 - Is an Expert required for the Gender Safeguarding Principles & Requirements?	The proposed project does not require an expert for the Safeguarding Principles Assessment as it is a Gender-Sensitive project and not a Gender-Responsive project. The project complies with all the requirements as per Gender Safeguarding Principles Assessment.
Question 4 - Is an Expert required to assist with Gender issues at the Stakeholder Consultation?	The proposed project did not require an expert for the Stakeholder Consultation meeting as it is a Gender-Sensitive project and not a Gender-Responsive project. In view of the Covid-19 outbreak and social distancing norms in Philippines and with reference to the GS Covid-19 Interim Measures, a delayed physical stakeholder meeting for 160MW Balaoi and Caunayan Wind Project was conducted on 11th July, 2022 as it was now safe to conduct. . The project developer had invited all the concerned community members as well as representatives from local, national authorities, GS NGO Supporters and GS representative for the stakeholder consultation meeting to ensure equal and

	effective participation from men and women and invites were sent out in a gender-sensitive manner. The project developer had used various types of invitation methods to invite all the relevant stakeholders, irrespective of their gender and to also ensure full participation of women and marginalized groups. Every participant was given an equal opportunity to communicate their thoughts openly during the discussion, regardless of discrimination on any basis.

SECTION E. SUMMARY OF LOCAL STAKEHOLDER CONSULTATION

The below is a summary of the 2 step GS4GG Consultation for monitoring purposes. Please refer to the separate Stakeholder Consultation Report for a complete report on the initial consultation and stakeholder feedback round.

E.1 Summary of stakeholder mitigation measures

During the stakeholder consultation meeting, the environmental, social and economic impacts of the project were very well taken by the participants and they all agreed that the project activity has no negative impacts and thus, no mitigation measures were suggested from any of the stakeholders.

The stakeholder feedback began on 20/07/2022 and lasted till 20/09/2022. The project developer did not receive any suggestions, inputs, comments from the stakeholders during this time period that will result in change in project design during the Stakeholder Feedback round.

E.2 Final continuous input / grievance mechanism

Method	**Include all details of Chosen Method (s) so that they may be understood and, where relevant, used by readers.**
Continuous Input / Grievance Expression Process Book (mandatory)	Process Book is available at BWPC Laoag Office – 2F Red Dot Collective Building, General Luna St. Corner Lagasca Street, Brgy 10, San Jose, Laoag City.
GS Contact (mandatory)	help@goldstandard.org

Other	The stakeholders can reach to Ms. Kimberly Rafael, Community Relation Officer for BWPC. The stakeholders can send their comments, suggestions, feedback by using any of the following means of communication: Email: kimberly.rafael@northluzon.com.ph Telephone: 077 7703721

APPENDIX 1 – SAFEGUARDING PRINCIPLES ASSESSMENT

Complete the Assessment below and copy all Mitigation Measures for each Principle into SECTION D above. Please refer to the instructions in the Guide to Completing this Form.

Assessment Questions/ Requirements	Justification of Relevance (Yes/potentially/no)	How Project will achieve Requirements through design, management or risk mitigation.	Mitigation Measures added to the Monitoring Plan (if required)
Principle 1. Human Rights			
1. The Project Developer and the Project shall respect internationally proclaimed human rights and shall not be complicit in violence or human rights abuses of any kind as defined in the Universal Declaration of Human Rights	No	The project does not include any activities which will result in violation of economic, social and cultural well-being of local communities. Therefore, project developer and project will respect internationally proclaimed human rights and shall not be complicit in violence or human rights abuses of any kind as defined in the Universal Declaration of Human Rights.	Not Required
2. The Project shall not discriminate with regards	No	The project will not discriminate with regards to participation	Not Required

to participation and inclusion		and inclusion based on caste, race, religion, gender, disability, language, origin or any other status.	
Principle 2. Gender Equality			
1. The Project shall not directly or indirectly lead to/contribute to adverse impacts on gender equality and/or the situation of women	No	The project will not engage into any such activities which will lead to discrimination towards women and will not cause any negative impact on socio-economic rights of women.	Not Required
2. Projects shall apply the principles of nondiscrimination, equal treatment, and equal pay for equal work	No	The project will always ensure that all employees will be treated equally in terms of work and employment opportunities, pay scale, workplace treatment, access to several benefits irrespective of their gender.	Not Required
3. The Project shall refer to the country's national gender strategy or equivalent national	No	The project will comply with country's international and national commitments to gender equality and will ensure	Not Required

commitment to aid in assessing gender risks 4. (where required) Summary of opinions and recommendations of an Expert Stakeholder(s)	No	women empowerment and gender equity at the workplace. Summary of opinions and recommendations of an expert stakeholder is required, only if the project is a "Gender-Responsive project activity". This project is a "Gender-Sensitive project activity".	Not Required
Principle 3. Community Health, Safety and Working Conditions			
1. The Project shall avoid community exposure to increased health risks and shall not adversely affect the health of the workers and the community	No	Occupational Health and Safety will always be taken care of during every stage of the project. In order to maintain the health and safety of employees at the workplace, employees will be given training in this respect from time to time. Appropriate measures will be taken to maintain the health of the community in close proximity to the project area. Hence, the project will comply with all the guidelines and	BWPC has its own Environment, Health and Safety policy. The policy framework consists of rules, guidelines, procedures, practices, responsibilities which the company and employees need to follow and comply with while performing all the operational activities at the workplace. In order to ensure safety of employees and environmental

		standards as per national Occupational Health and Safety Standards and International Labour Organization fundamental conventions.	protection, company will give necessary health and safety up-to-date training to all of its employees. BWPC will conduct annual audit to review EHS plan and policy and besides this, the management will conduct audits periodically so as to evaluate and implement appropriate measures.
Principle 4.1 Sites of Cultural and Historical Heritage			
Does the Project Area include sites, structures, or objects with historical, cultural, artistic, traditional or religious values or intangible forms of culture?	No	The project area is a part of public forestland and does not include sites, structures, or objects with historical, cultural, artistic, traditional or religious values or intangible forms of culture.	Not Required
Principle 4.2 Forced Eviction and Displacement			
Does the Project require or cause the physical or economic relocation of peoples (temporary or permanent, full or partial)?	No	The project area is a public forestland and does not require or cause physical or economic relocation of people.	Not Required
Principle 4.3 Land Tenure and Other Rights			

a. Does the Project require any change, or have any uncertainties related to land tenure arrangements and/or access rights, usage rights or land ownership? b. For Projects involving land use tenure, are there any uncertainties with regards to land tenure, access rights, usage rights or land ownership?	No No	The project developer holds All the rights to the project land and does not require any change to land tenure arrangements. Not applicable as the project does not involve any change to land tenure arrangements.	Not Required Not Required
Principle 4.4 – Indigenous people			
Are indigenous peoples present in or within the area of influence of the Project and/or is the Project located on land/territory claimed by indigenous peoples?	No	Yes, indigenous people live around the project area. BWPC has already entered into an MOA with the indigenous peoples.	Not Required
Principle 5. Corruption			
1. The Project shall not involve, be complicit in or inadvertently contribute to or reinforce	No	The project will not be a part of any such activities which constitute or may lead to corruption and BWPC has its	Not Required

corruption or corrupt Projects		own strict rules and policies against corruption and bribery.	
Principle 6.1 Labour Rights			
1. The Project Developer shall ensure that all employment is in compliance with national labour occupational health and safety laws and with the principles and standards embodied in the ILO fundamental conventions	No	Occupational Health and safety will always be taken care of during every stage of the project. In order to maintain the health and safety of employees at the workplace, employees will be given training in this respect from time to time. Hence, the project will comply with all the guidelines and standards as per national Occupational Health and Safety Standards and International Labour Organization fundamental conventions.	The employment will be in compliance with the national employment legislations and occupational health and safety laws will be followed in order to maintain the health and safety of employees at the workplace.
2. Workers shall be able to establish and join labour organisations	No	Employees will not be deprived of their right to self-organization and thus, will not be restricted from establishing and joining labour organizations as sanctioned by the Labour Code of Philippines.	

3. Working agreements with all individual workers shall be documented and implemented and include: a) Working hours (must not exceed 48 hours per week on a regular basis), AND b) Duties and tasks, AND c) Remuneration (must include provision for payment of overtime), AND d) Modalities on health insurance, AND e) Modalities on termination of the contract with provision for voluntary resignation by employee, AND f) Provision for annual leave of not less than 10 days per year, not including sick and casual leave.	No	As per Bureau of Working Conditions, Department of Labour and Employment, the working agreements of every employee will be properly documented and will cover all the aspects of legal employment practices as well as labour relations. Besides this, employee handbook will also be given to state responsibilities and rights of employees.	

4. No child labour is allowed (Exceptions for children working on their families' property requires an Expert Stakeholder opinion)	No	The project will not employ children during any phase of project for any type of work.	
5. The Project Developer shall ensure the use of appropriate equipment, training of workers, documentation and reporting of accidents and incidents, and emergency preparedness and response measures	No	Occupational Health and Safety will always be taken care of during every stage of the project. In order to the maintain health and safety of employees at the workplace, employees will be given training in this respect from time to time.	
Principle 6.2 Negative Economic Consequences			
1. Does the project cause negative economic consequences during and after project implementation?	No	The project will not cause negative economic consequences during and after project implementation. It will in fact provide permanent employment opportunities to locals including women.	Not Required

Principle 7.1 Emissions			
Will the Project increase greenhouse gas emissions over the Baseline Scenario?	No	The project will reduce GHG emissions compared to the baseline scenario.	Not Required
Principle 7.2 Energy Supply			
Will the Project use energy from a local grid or power supply (i.e., not connected to a national or regional grid) or fuel resource (such as wood, biomass) that provides for other local users?	No	The project activity is a grid-connected wind power generation project which will supply electricity to the national grid and will not use energy from a local grid or power supply (i.e., not connected to a national or regional grid) or fuel source (such as wood, biomass).	Not Required
Principle 8.1 Impact on Natural Water Patterns/Flows			
Will the Project affect the natural or pre-existing pattern of watercourses, ground-water and/or the watershed(s) such as high seasonal flow variability, flooding potential, lack of aquatic connectivity or water scarcity?	No	The project is a wind energy project and the project area is a part of a public forestland. It will neither contaminate nor restrict the natural flow of any surface and sub-surface watercourses and has already obtained Environmental Compliance Certificate from the	Not Required

		Department of Environment and Natural Resources, Philippines.	
Principle 8.2 Erosion and/or Water Body Instability			
a. Could the Project directly or indirectly cause additional erosion and/or water body instability or disrupt the natural pattern of erosion?	No	The project may cause soil erosion during the construction phase but mitigation steps such as engineering measures (construction of riprap and stone masonry) and bioengineering measures (planting of vetiver grass and installation of cocologs and coconets) will be taken.	Not Required
b. Is the Project's area of influence susceptible to excessive erosion and/or water body instability?	No	The project may cause soil erosion during the construction phase but mitigation steps such as engineering measures (construction of riprap and stone masonry) and bioengineering measures (planting of vetiver grass and installation of cocologs and coconets) will be taken to reduce the impact.	Not Required

Principle 9.1 Landscape Modification and Soil			
Does the Project involve the use of land and soil for production of crops or other products?	No	The contract area of the project is a public forestland and the project will harness wind energy to generate electricity and will not involve use of land and soil for production of crops or other products.	Not Required
Principle 9.2 Vulnerability to Natural Disaster			
Will the Project be susceptible to or lead to increased vulnerability to wind, earthquakes, subsidence, landslides, erosion, flooding, drought or other extreme climatic conditions?	No	The project will generate electricity by using wind energy, a clean, sustainable and renewable energy source. Due to country's geographical location, it is highly exposed to natural disasters but the project contributes towards low-carbon economic development activities and therefore, will not be susceptible to or lead to increased vulnerability to wind, earthquakes, subsidence, landslides, erosion, flooding, drought or other extreme climatic conditions.	Not Required
Principle 9.3 Genetic Resources			

Could the Project be negatively impacted by or involve genetically modified organisms or GMOs (e.g., contamination, collection and/or harvesting, commercial development, or take place in facilities or farms that include GMOs in their processes and production)?	No	The aim of the project is to harness wind energy to generate electricity and supply it to the grid. Hence, the project will neither be negatively impacted by or involve genetically modified organisms or GMOs.	Not Required
Principle 9.4 Release of pollutants			
Could the Project potentially result in the release of pollutants to the environment?	No	The project is a wind energy project, so it will not result in release of pollutants to the environment. The project will have a temporary impact on the environment only during the construction phase due to soil disruption and respective measures will be taken to minimize the impact.	Not Required
Principle 9.5 Hazardous and Non-hazardous Waste			
Will the Project involve the manufacture, trade, release, and/ or use of hazardous and non-hazardous chemicals and/or materials?	No	The aim of the project is to harness wind energy to generate electricity and supply it to the grid. Therefore, the project will not involve	During operation phase of the project, the waste will be collected, segregated and disposed of as per local guidelines and waste

		manufacture, trade, release, and/ or use of hazardous and non-hazardous chemicals and/or materials.	management laws, through an authorized waste handler.
Principle 9.6 Pesticides & Fertilisers			
Will the Project involve the application of pesticides and/or fertilisers?	No	The aim of the project is to harness wind energy to generate electricity and supply it to the grid. Therefore, the project will not involve application of pesticides and/or fertilizers.	Not Required
Principle 9.7 Harvesting of Forests			
Will the Project involve the harvesting of forests?	No	The project is located on a public forestland and the aim of the project is to harness wind energy to generate electricity and supply it to the grid. Therefore, the project will not involve harvesting of forests.	Not Required
Principle 9.8 Food			
Does the Project modify the quantity or nutritional quality of food available such as through	No	The aim of the project is to harness wind energy to generate electricity and supply it to the grid. Therefore, the	Not Required

crop regime alteration or export or economic incentives?		project will not modify the quantity or nutritional quality of food available such as through crop regime alteration or export or economic incentives.	
Principle 9.9 Animal husbandry			
Will the Project involve animal husbandry?	No	The aim of the project is to harness wind energy to generate electricity and supply it to the grid. Therefore, the project will not involve animal husbandry.	Not Required
Principle 9.10 High Conservation Value Areas and Critical Habitats			
Does the Project physically affect or alter largely intact or High Conservation Value (HCV) ecosystems, critical habitats, landscapes, key biodiversity areas or sites identified?	No	The project area is not classified as a critical habitat and being a renewable energy project, it does not affect or alter any HCV ecosystems, critical habitats, landscapes, key biodiversity areas or sites.	Not Required
Principle 9.11 Endangered Species			
a. Are there any endangered species identified as potentially being present within the Project boundary	No	As per the Biodiversity Assessment Report, the project site is majorly dominated by native and endemic species	Not Required

(including those that may route through the area)?		(having least concern status by IUCN and DENR) of flora and fauna. A very small percentage of threatened species of flora and fauna having (critically endangered/ endangered/vulnerable) status are present within the project area. Due to pre-existing practices such as ecosystem services of flora, collection and hunting of fauna, habitat destruction and conversion to agricultural land/field, has resulted in threatened status of some of the species from both categories. Although the project causes some temporary disturbance during the construction period for which BWPC will be taking mitigation measures to reduce the impact. The report states that the project will neither adversely affect the threatened species nor their habitat and BWPC's biodiversity conservation and	

		management efforts will turn out to be beneficial for the ecosystem.Also, the biodiversity assessment states that project area is not a flyway for migratory birds as there are no wetlands present within the project site for birds to glean and search for food. To reduce bird mortality among the native and endemic bird species, effective measures will be taken to reduce bird mortality.	
b. Does the Project potentially impact other areas where endangered species may be present through transboundary affects?	No	The project will be only limited to the boundary of the Project area and hence, will not impact other areas where endangered species may be present through transboundary affects.	Not Required

APPENDIX 2- CONTACT INFORMATION OF PROJECT PARTICIPANTS

Organization name	Bayog Wind Power Corp.
Registration number with relevant authority	WESC No. 2010-02-038
Street/P.O. Box	General Luna Street
Building	Red Dot Collective Building
City	San Jose, Laoag City
State/Region	Ilocos Norte
Postcode	2900
Country	Philippines
Telephone	+6377 7703721
E-mail	mark.tagum@northluzon.com.ph
Website	
Contact person	Mark Anthony Tagum
Title	President
Salutation	Mr.
Last name	Tagum
Middle name	
First name	Mark Anthony
Department	
Mobile	+63917 7088067
Direct tel.	
Personal e-mail	mark.tagum@northluzon.com.ph

APPENDIX 3- LUF ADDITIONAL INFORMATION

Not Applicable

APPENDIX 4-SUMMARY OF APPROVED DESIGN CHANGES

Not Applicable

Revision History

Version	Date	Remarks
1.2	14 October 2020	Hyperlinked section summary to enable quick access to key sections Improved clarity on Key Project Information Inclusion criteria table added Gender sensitive requirements added Prior consideration (1 yr rule) and Ongoing Financial Need added Safeguard Principles Assessment as annex and a new section to include applicable safeguards for clarity Improved Clarity on SDG contribution/SDG Impact term used throughout Clarity on Stakeholder Consultation information required Provision of an accompanying Guide to help the user understand detailed rules and requirements
1.1	24 August 2017	Updated to include section A.8 on 'gender sensitive' requirements
1.0	10 July 2017	Initial adoption

ANNEX I – ASSUMPTIONS SHEET

Technology			Source	Variation for

				sensitivity analysis
Wind turbine model		Siemens Gamesa		
Number of turbines		32	*Feasibility Study Report, Service Maintenance and Availability Agreement*	
Turbine capacity	MW	5		
Total Capacity	MW	160	*Feasibility Study Report, Service Maintenance and Availability Agreement*	
Investment Decision		6-Oct-20	*Secretary's Certificate*	
Expected COD		15th December 2022		
Years of Operation	Years	20	*Schedule G4*	
Production				
Electricity production (P75)	MWh	532,300	*Energy Yield Assessment Report and BWPC Wind Farm Consumption Report*	0%
Net Electricity Supplied to the Grid	MWh	532,300	Net Electricity Supplied to the Grid	
Economic Assumptions				
Exchange rate versus USD	Peso	49.5	*Project Financial Model*	
Inflation Rate		0%	*Assumed for nominal analysis*	
Electricity Tariff (WESM)	Peso/kWh	3.65	*Feasibility Study Report*	0%
Line Rental	Peso/kWh	0.3	*Based on BWPC's development experience*	0%
Net Tariff (WESM)	Peso/kWh	3.35	*Project Financial Model*	
Category		**Peso ('000)**	*Project Financial Model*	

Total WTG Supplier Scope		49,08,712		
Total Inland Transportation		7,90,197		
Power Curve Testing		25,296		
Total Owner's Engineer Fees		9,158		
Batching Plant		37,861		
Total Substation and Electrical		8,87,646		
Warehouse		20,618		
Total Civil Works		15,26,539		
Transmission Line		4,09,116	*Based on BWPC’s development experience*	
TOTAL PRIMARY COSTS		**86,15,143**		
Development Costs		4,59,595	*Based on BWPC’s development experience*	
VAT input		7,69,330		
Construction Insurance		1,14,727		
Other Project Company Expenses		2,78,970		
IP Payments		45,860		
EPC Cost Saving Bonus		2,10,375		
Advance Payment to SGRE (Service)		18,350		
Insurance Adviser		5,940		
Initial working capital		-		
TOTAL OTHER COSTS		**19,03,147**		

Upfront fees and DST		93,342		
Interest during construction		-		
Commitment Fees		-		
DSRA funding requirement at COD		-		
Security Reg/Letter of Credit		-		
TOTAL FINANCING COSTS		**93,342**		
Contingency		7,95,875		
TOTAL PROJECT COSTS		1,14,07,507		0%
Depreciation				
Salvage value	*Peso*	0	*Based on BWPC's development experience*	
Capital Costs (Equipment Costs)			*Project Financial Model*	
% of costs Depreciable		100%		
Depreciation Basis		St. Line		
Years		20	*Schedule G4*	
Development Cost				
% of costs Depreciable		100%		
Depreciation Basis		St. Line		
Years		20		
Tax Assumptions			*Project Financial Model*	
Income Tax (1-3 years)		0%	*Required by law*	

Income Tax rate (4-6 years)		0%		
Income Tax rate (7-20 years)		10%		
Operating Costs ('000) Pesos			*Project Financial Model*	
Personal Expenses		47,000		
Land Rent		3,700		
Rental Increase per Year		10%	*Required by law*	
SGRE Maintenance		72096	*2,253 per wtg/yr with escalation per year*	
Spares		11042	*every 5 years*	
Government Share		1%	*Gross Profit*	
Local Business Tax Rate		0.45%	*of revenues*	
ER 1-94		0.010	*per kWh*	
Market Fees		0.010	*per kWh*	
CSR		10,219		
Utilities Charges (WESM and NGCP)		14,000		
Security Cost		9,000		
Service Vehicle		2,300		
Professional Fees (O&M Consultant, Legal, Wind analysis, Accounting, SAP)		10,400		
Retirement Fund		1.5	*months of personnel expenses*	
Office expenses		13,000		
Training		2,000		

Insurance ('000)			*Project Financial Model*	
All risks		0.50%	*of primary costs*	
3rd Party liability, PHP		1500		
Bsuiness Interruption		0.70%	*of Gross Profit*	
Real Property Tax ('000)			*Project Financial Model*	
Primary costs		94,11,018		
Real Property Tax Rate		1.5%	*Local regulation*	
RPT Basis as % of Depreciated Primary Costs		75%	*Local regulation*	